EHIGNITE CHALLENGE · PHASE 1 CONCEPT & DESIGN

EHI Atlas

A patient-owned, portable evidence workspace that turns fragmented Single Patient EHI exports into source-backed summaries, charts, Q&A, CLI review, and clinician-ready handoffs.

SUBMISSION TYPE	PRIMARY SCENARIO	ADDITIONAL SCENARIOS	STATUS
Phase 1 narrative	Integration across settings	Clinical customization · Interactive tools	Working prototype + design

We make patient records portable and understandable before making them intelligent: fragmented EHI exports become a patient-owned evidence workspace that humans can inspect, tools can query, and AI can safely assist with.

The problem

Patients increasingly have a right to receive their health information, but the practical output is often a folder of files: FHIR bundles, C-CDAs, PDFs, portal exports, scanned reports, and lab documents. The data is available, but not usable.

- What changed since the last visit?
- Which medications are active now?
- What information is missing before a second opinion, surgery, or specialist visit?
- Which source supports each claim?

The solution

EHI Atlas prepares each source, maps facts into FHIR-compatible resources, preserves provenance, harmonizes across sources, then exposes the result through summaries, charts, Q&A, and portable handoffs.

Readable · Usable · Actionable · Standards-aligned · AI-ready

One-sentence positioning

EHI Atlas turns EHI availability into EHI usability. It is not a PDF chatbot. It is an interpretable, portable record workspace with a structured clinical evidence layer underneath and multiple interfaces on top: web UI, CLI review, exports, charts, and AI assistance.

The model is not the product. The durable product is the patient-owned evidence workspace; models and agents are interfaces on top.

PROBLEM ALIGNMENT

Why raw EHI exports fail users

Current exports are often technically compliant but cognitively inaccessible. They move data, but they do not answer user questions. A patient, caregiver, or receiving clinician must still reconcile duplicated facts, interpret codes, sort administrative noise from clinical signal, and determine whether the record is complete enough to act.

Fragmented

Records arrive across portals, labs, specialists, payers, PDFs, C-CDAs, and FHIR bundles.

Document-centric

Most artifacts preserve the source document but do not create a reusable computational record.

Hard to trust

AI summaries are unsafe if they cannot cite the source facts and disclose missing information.

Challenge scenarios addressed

Scenario	How EHI Atlas addresses it
Integration Across Settings	Ingests FHIR bundles, C-CDA/CDA, PDFs, lab reports, and portal exports into one FHIR-compatible fact graph with source provenance.
Customization for Clinical Domains	Supports domain-specific evidence packets for pre-op review, medication reconciliation, second opinions, caregiver handoffs, and future specialties.
Interactive Patient Tools	Provides cited Q&A over the patient's own harmonized record, with missing-information flags instead of unsupported answers.

Design target

> **The right five facts in thirty seconds — with a path back to the source.**

Atlas prioritizes actionability: summaries and charts first, evidence drilldown second, raw artifacts always available. This keeps the user out of database browsing while preserving auditability.

ARCHITECTURE

A portable evidence layer for interpretable EHI

FHIR is not the user interface and it is not the final product. A FHIR-compatible structure is the common evidence layer that makes fragmented records consistent, auditable, easier to interpret, and portable across tools.

1. EHI Sources
FHIR R4, C-CDA, PDFs, lab reports, portal exports, patient context

2. Source Intake
Classify, prepare, extract, and validate readiness

3. Fact Graph
Observation, Medication, Condition, Encounter, DocumentReference, Provenance

4. Trust Layer
Deduplicate, harmonize, validate, detect conflicts, attach citations

5. Outputs
Summaries, charts, cited Q&A, handoffs, missing-info flags

Parse once. Structure once. Validate once. Cite forever.

Why this is better than direct document summarization

Direct document AI	EHI Atlas pattern
LLM rereads raw files on each request.	System retrieves typed, validated facts for the question.
Prompt behavior depends on document layout and context window.	Agent tools operate over stable clinical concepts.
Citations are difficult and often approximate.	Each fact retains source document, page/location, method, and provenance.
Hard to reuse across sources.	New inputs become adapters into the same clinical substrate.

Agent tool examples

```
get_recent_labs(loinc, window) · get_active_medications(rxnorm_class) ·
get_condition_history(problem) · compare_sources(fact_id) · get_source_provenance(fact_id) ·
build_patient_summary(audience, purpose)
```

The model becomes transportable because it reasons through tools over standardized facts, not custom prompts tied to one PDF or C-CDA layout.

PRODUCT WORKFLOW

From source files to a usable record

Source Intake

Add files, classify each source, prepare for harmonization.

File	Type	Prep state
portal-export.json	FHIR	Ready
lab-report.pdf	PDF	Extracted
ccd.xml	C-CDA	Validate

Every source has a preparation log and downstream provenance trail.

Harmonized Record

Canonical record built from prepared sources.

Sources: 3 Canonical facts: 206 Shared facts: 26

Open review: 1

Fact	Sources	Status
Creatinine	2	Shared
Metformin	1	Unique
Albumin	2	Shared

Patient Context

Patients and caregivers add goals, symptoms, concerns, care tasks, and questions that formal EHI may miss.

Formal record + human context = better handoff.

Publish Chart

Create a scoped chart snapshot for a second opinion, specialist visit, pre-op review, caregiver handoff, or payer workflow.

Share the right evidence, not the entire raw archive.

Product principle

Source-specific at the edge, standards-aligned at the core. FHIR, C-CDA, CDA, and PDFs remain important inputs. The durable product layer is the harmonized, provenance-backed clinical fact graph.

PROTOTYPE STATUS

What has been built and tested

FHIR corpus + clinical workspace

Prototype explorer over a synthetic Synthea R4 corpus: approximately 1,180 patients and 527,000+ resources. Surfaces include overview, safety panel, interactions, timeline, care journey, conditions, procedures, immunizations, clearance, and assistant.

Data Lab + methodology

Reviewer-facing environment explaining FHIR resource types, data definitions, coverage, pipeline stages, and trust/interpretability gates. This makes the methodology inspectable, not hidden.

Assistant context engineering

The assistant builds query-filtered clinical context from safety flags, medication classes, ranked conditions, allergies, encounters, and other facts instead of injecting raw bundles into the model.

PDF extraction lab

PDF-to-FHIR pipeline with document-context pass, focused per-resource extraction passes, FHIR Bundle output, source locators, and evaluation against gold-standard outputs where available.

Local/specialized model testing

Recent work added a MedGemma 4B / Ollama pipeline for bounded labs/vitals extraction. This is intentionally benchmarked as a candidate path rather than assumed to be sufficient for all clinical extraction. The architecture is model-flexible: frontier models can handle complex narrative extraction while local or specialized models can be tested for repetitive tabular tasks.

Area	Current evidence	Phase 2 next step
FHIR corpus	Synthetic R4 patient corpus and clinical/data lab surfaces.	Expand multi-source evaluation and reviewer workflows.
PDF extraction	Pluggable pipelines emit FHIR Bundle outputs.	Benchmark precision/recall/cost/latency by resource type.
Local models	MedGemma 4B installed and smoke-tested for labs/vitals.	Compare against cloud gold outputs and improve cropping/OCR prompts.

TRUSTWORTHY AI DESIGN

AI assistance over prepared, interpretable evidence

The product should make the record usable before AI enters the loop. When the assistant is used, it receives a bounded evidence packet assembled from validated clinical facts. This keeps responses faster, more explainable, and less likely to drift.

Intent
Classify the user question and clinical purpose.

Retrieve
Pull relevant facts, source refs, and risk signals.

Reason
Ask the LLM to explain from the bounded evidence packet.

Return
Cited answer, missing information, next actions.

Example assistant answer pattern

Question: “Is this patient safe for surgery this week?”

Evidence packet: 15 facts · safety flags, active medications, recent encounters, missing labs · source citations attached

NSAID active · hold 3–5 days | Metformin · peri-op protocol | No anticoagulant found | Missing recent CBC/CMP

Answer: Not yet cleared. Address medication holds and obtain missing baseline labs. See source records for each claim.

Portable tooling surface

Atlas should not trap the patient's record inside one web application. The same evidence workspace can support command-line review, deterministic charting, markdown/JSON/FHIR exports, local models, frontier models such as Claude, and future clinical agent systems. This portability is part of patient ownership: the patient can carry usable evidence across tools rather than starting over with each new application.

AI guardrails

- **Deterministic-first harmonization:** matching, deduplication, source contribution, and provenance are explicit and testable.
- **Evidence minimization:** the LLM receives the minimum necessary facts for the question, not unlimited raw record access.
- **Missing-information disclosure:** the assistant should say what it does not know.
- **Source-backed outputs:** important claims link back to the underlying clinical facts and documents.

The point is not that one Atlas assistant beats every model. The point is that a patient-owned evidence workspace makes capable models and tools better, safer, and more traceable.

FEASIBILITY, SCALABILITY, PRIVACY

Technical feasibility and scalability

Atlas separates source-specific ingestion from standards-aligned reasoning. Each input format gets an adapter, and every adapter targets the same FHIR-compatible clinical fact graph.

Layer	Function	Scalability logic
Ingest	Archive and classify raw files.	Add new source types without changing downstream views.
Extract/adapt	Convert FHIR, C-CDA, PDFs, and reports into candidate facts.	Pipeline registry supports multiple extraction strategies.
Harmonize	Deduplicate, validate, code, and connect facts to provenance.	Deterministic logic can be tested across corpora and source mixes.
Reason	LLM tools retrieve evidence packets for summaries, Q&A, and charts.	Same tools work across sources once facts are normalized.

Privacy and compliance posture

The Phase 1 prototype uses synthetic Synthea data wherever possible and keeps private/PHI-like test artifacts out of source control. The architecture is designed for HIPAA-aware deployment patterns such as tenant isolation, customer-controlled environments, audit logging, and scoped patient-controlled sharing.

Minimum necessary context

The LLM receives focused evidence packets, not unrestricted raw artifact access.

Provenance by design

Facts retain source trails to avoid orphaned clinical claims.

Private inference option

Local/specialized model paths can support privacy, cost, and deployment flexibility.

Standards alignment

FHIR R4 **US Core where applicable** **SMART App Launch-ready future path** **C-CDA/CDA as source inputs** **Provenance-centered design**

IMPACT AND TEAM

Potential impact

Patients and caregivers

Understand the record, ask better questions, see trends, identify gaps, and bring a coherent summary to new care settings.

Clinicians and care teams

Review high-signal summaries with source references, spot conflicts, and generate handoffs without trusting a black-box summary.

Interoperability ecosystem

Convert EHI movement into EHI usability through a standards-aligned evidence layer that can power multiple applications.

Rubric alignment

Criterion	EHI Atlas response
Relevance & problem alignment	Directly targets unusable Single Patient EHI exports and creates readable, actionable summaries.
Integration & scaling	Adapter model; FHIR-compatible core; multi-source harmonization and source contribution tracking.
Interpretability & ease of use	Five-facts design target; patient/clinician summaries; charts; citations; Data Lab methodology.
Privacy/security/compliance	Synthetic demo data; evidence minimization; patient-controlled sharing; private deployment path.
AI innovation	Portable evidence workspace, context packets, provenance-backed reasoning across agent systems, local model evaluation.

Submitting individual / team

Blake Thomson brings healthcare data strategy and business development experience from Cedars-Sinai, where his work focuses on claims data, referral ecosystems, provider networks, and translating complex healthcare data into operational strategy. This submission combines healthcare domain understanding with hands-on prototyping in FHIR data modeling, agentic workflows, PDF extraction, clinical context engineering, local model evaluation, and patient-facing health information design.

> **The goal is not AI novelty in isolation. The goal is to make patient-controlled health information practically useful, explainable, and portable.**